

Physics-informed machine learning for identification of preferential flow paths and performance forecasting of subsurface carbon storage and utilization

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ABSTRACT

Monitoring the CO₂ plume movement in the subsurface is essential for safety and storage integrity during CO₂ storage and utilization. During field-scale CO₂ EOR, routine well-wise injection/production data contain significant information which can be used for closed-loop reservoir management. Traditional physics-based numerical reservoir simulation can be computationally prohibitive for short-term decision cycles, and it requires detailed geologic models. As an alternative, reduced physics models provide an efficient simulator-free workflow but often have a limited range of applicability. Pure machine learning models lack physical interpretability and fail to provide process-based insights, resulting in limited predictive power. To address these challenges, we propose hybrid models, combining machine learning and physics-based approach, for rapid production forecasting and reservoir connectivity characterization using routine injection/production and pressure data collected during CO₂ EOR operation.

We combine reduced physics models into a neural network architecture by utilizing two different approaches. In the first approach, the reduced physics model is used for pre-processing to obtain approximate solutions that feed into a neural network as input. This physics-based input feature can reduce the model complexity and provide significant improvement in prediction performance. In the second approach, physics-informed neural network (PINN) is applied. The residual terms are augmented in the neural network loss function using a physics-based regularization that relies on the governing partial differential equations (PDE). Reduced physics models are used for the governing PDE to enable efficient neural network training.

Our proposed hybrid models are first validated using a benchmark reservoir simulation case and then applied to a field case to show the robustness and efficacy of the method. The hybrid models are shown to provide superior prediction performance than pure machine learning models in terms of multiphase production rates. Specifically, the trained PINN model satisfies the reduced physics system, providing inter-well connectivity in terms of well-flux allocation. The flux allocation estimated from the hybrid model was compared with streamline-based flux allocation, and reasonable agreement was obtained for both the benchmark case and the field case.

1. Introduction

CO₂ storage and utilization are critical technologies to minimize greenhouse gas emission and control global warming (Michael et al., 2010; Sharma, 2011; Viebahn et al., 2015; Zhou et al., 2016; Aminu et al., 2017). The injected CO₂ needs to be securely isolated from the surrounding environment because CO₂ leakage could cause contamination of groundwater and endanger human health (Benson and Myer,

2003; Wilkin and DiGiulio, 2010). Near real-time monitoring and verification of the CO₂ plume movement is essential to understand CO₂ migration in the reservoir for risk management and operational decision making.

Carbon dioxide enhanced oil recovery (CO₂-EOR) has been a successful commercial activity in the U.S. since 1972, when it first started in the Permian Basin SACROC Field in Texas (Núñez-López and Moskal, 2019). Oil field development is typically carried out in two or three

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recovery stages, primary recovery, waterflooding, and the EOR is often considered a tertiary recovery (Núñez-López and Moskal, 2019). About one-third of the resource base has been recovered after primary and secondary oil production, which means that a significant target of oil is still trapped in the subsurface (Ferguson et al., 2010). A large portion of the remaining oil is recoverable through CO₂-EOR application (Ferguson et al., 2010). Furthermore, CO₂-EOR has been an attractive and commercially established technique to store greenhouse gas underground. Based on the international goal of climate change mitigation, it is clear that large-scale emission reduction technologies such as CCS will be necessary to meet the net-zero emission goal (Núñez-López and Moskal, 2019). In fact, current operational experience shows that more than 80 % of the injected CO₂ remains geologically trapped (Melzer, 2012). The primary risk that is higher in EOR than CCS in saline aquifer is the risk of leakage through legacy wells. However, if operators take the necessary precautions, the benefits can be significant in terms of carbon utilization and storage (Núñez-López and Moskal, 2019).

Several history matching and field optimization methods that utilizes full-physics numerical simulations for CO₂-EOR have been studied. Olalotiti-Lawal et al., 2019b applied a hierarchical history matching approach for geologic model calibration by incorporating available pressure and multiphase production data in a large-scale post combustion CO₂-EOR project. A CO₂ water-alternating-gas (WAG) pilot was then conducted in the field. Olalotiti-Lawal et al., 2019a incorporated pilot operation data by the hierarchical history matching workflow to obtain a reliable dynamic model. Then, CO₂-EOR operation optimization is conducted on the calibrated pilot dynamic model, which gives guidance for field wide CO₂-EOR operation. Yao et al. (2021) applied a hierarchical history matching workflow to a large-scale CO₂ storage test site located at the Northern Niagaran Pinnacle Reef Trend in Michigan. Their study integrated field monitoring data, including distributed pressure and distributed temperature sensing (DTS) data. Tanaka et al. (2017) developed an efficient streamline-based rate allocation optimization framework for maximizing NPV. The method was successfully applied to a CO₂-EOR reservoir case. Although full physics numerical simulations for CO₂-EOR cases are computationally intensive due to the complex physics and dimensionality of the reservoir models, it can account for the underlying physics of the flow problems. Also, robust history matching and optimization workflows are well-established. Tanaka et al. (2014) developed a 3D 3-phase compositional streamline simulator incorporating the gravity and capillarity effects and applied to a field-scale CO₂ injection case, demonstrating improved computational efficiency compared to full-physics 3D simulations. Tracer methods can also directly reflect the fluid migration paths through high-permeability channels. However, it can have high implementation cost and often requires long testing period (Patidar et al., 2022).

For field-scale problems, it can be challenging to obtain reliable reservoir models, and history matching process can be computationally prohibitive, especially if we rely on the full physics numerical simulations with 3D geo-cellular models. To mitigate these challenges, extensive studies have been conducted on reduced physics-based data driven models for efficient history matching and optimization workflows. The data-driven models can be calibrated solely based on the historical well measurements without relying on the prior geologic information so that the calibration process can be computationally efficient. The Capacitance Resistance Model (CRM) was introduced as an efficient reduced physics model, which treats injection rates and pressure data as input signals and total production rates as the output signal, and they are correlated by using a simple analytical equation (Albertoni and Lake, 2003; Al-Yousef, 2006; Yousef et al., 2006). The history matched models provide physically interpretable parameter's such as injection allocation factor, time delay constant, and productivity index. The CRM was further extended to account for multi-well case by considering three different control volume cases using the concept of superposition in time; CRMT: one control volume for the entire field, CRMP: one control volume for each producer, and CRMIP: one control volume for each

injector-producer pair (Sayarpour et al., 2007; Sayarpour, 2008). The CRM has been successfully applied to CO₂-EOR and CO₂ storage problems (Sayarpour, 2008; Nguyen, 2012; Laochamroonvorapongse et al., 2014; Tao and Bryant, 2015; Yousefi et al., 2021). For pursuing the efficiency and simplicity of the implementation in CRM, it assumes simplified underlying physics formulation, such as constant productivity indices, constant injection allocation factor, and single control volume (tank) between producer and injector pair. Zhao et al. (2016) developed a reduced physics model called inter-well numerical simulation (INSIM) model based on the concept of streamtubes, which eliminates some of the simplifying assumptions in CRM and allows for transient productivity index and dynamic allocation factors. The framework was extended by utilizing a front-tracking approach to improve the multi-phase production estimation (Guo et al., 2017, 2018), which is called INSIM-FT. Several different types of flow network models have been proposed to consider more complex physics (Lerlertpakdee et al., 2014; Lutidze, 2018; Ren et al., 2019). The flow network models construct a network that connects the wells or completions by using a series of 1D grid cells so that the number of grid cells and computational time can be significantly reduced. The history matching parameters, including grid block pore volume and transmissibility are calibrated solely based on the historical well measurements. The reservoir graph network (RGNet) model constructs the flow network system based on the concept of Diffusive Time of Flight (DTOF) (Fujita et al., 2016; Zhang et al., 2016), in which each well or completion can have multiple 1D grid blocks, and connections between wells or completions are represented by a single transmissibility (Sankaran and Sun, 2020; Sun and Sankaran, 2021; Guo and Sankaran, 2022). Many of the flow network models are linked with commercial reservoir simulator, so that they do not need to develop nonlinear solver from scratch, which requires significant effort.

Recent rapid advancement in machine learning have offered another class of efficient data-driven approach for CO₂ injection and EOR problems. Chen et al. (2021) and Sen et al. (2022) applied special type of Recurrent Neural Network (RNN) (Hopfield, 1982; Rumelhart et al., 1986) called Statistical Recurrent Unit (SRU), developed by Oliva et al. (2017), to identify the reservoir connectivity between injectors and producers by correlating the routine well measurements. This framework is successfully applied to a large-scale CO₂-EOR reservoir field. Convolutional Neural Network (CNN) is a class of neural network architecture that is utilized for subsurface image reconstruction/prediction problems. Nagao et al., 2023b developed an efficient and scalable CNN-based framework for predicting the CO₂ plume evolution based on available monitoring data, such as distributed pressure and distributed temperature sensing (DTS) data. This CO₂ plume visualization framework is successfully applied to large-scale CCS project of depleted oil reservoir and saline aquifers (Nagao et al., 2022a,c; Nagao et al., 2023b; Nagao et al., 2024b, Takuto Sakai et al., 2024). The machine learning approaches can provide efficient model calibration and optimization workflow with wide range of applicability. However, it also has several drawbacks. First, it requires sufficient datasets to properly learn the correlations between data, which can be too expensive to obtain and may not be available for field scale applications. Second, pure machine learning models are less interpretable compared to the physics-based models, as the calibrated parameters in pure machine learning models do not have physical significance.

In order to mitigate the shortcomings of the pure machine learning models and reduced physics models, we utilize physics informed hybrid machine learning approaches, where physics-based models and machine learning models are combined into a single framework. Three different types of hybrid machine learning approaches have been proposed, including physics-based preprocessing, physics-based architecture, and physics-based regularization (Rai and Sahu, 2020; Willard et al., 2020). Physics-based preprocessing improves the robustness of the machine learning models by augmenting input features that are given from physics-based solutions. Nagao et al., 2022b incorporated this approach with dynamic mode decomposition (DMD) and sparse identification of

Table 1

Summary of advantages and limitations of existing methods for reservoir connectivity identification and production forecasting.

Existing Methods	Advantages	Limitations
Tracer test	Reflects actual fluid flow in reservoirs	• High cost • Long testing time
Full-physics simulation	Consideration of complex physics	High computational cost
Reduced physics model	• Computational efficiency • Interpretability	• Assumptions of simplified physics • Limited range of applicability
Pure machine learning	Computational efficiency	• Large amount of data is required • Lack of interpretability

nonlinear dynamics (SINDy) for reservoir surveillance problems, in which they estimate and track the productivity indices in near real time based on the available well measurements. [Molinari and Sankaran \(2021\)](#) investigated the robustness of the different PINN hybrid machine learning approaches for predicting the bottom-hole pressure by combining physics-based flow correlations with data-driven models to capture the field bottomhole pressure estimation. The physics-informed neural network (PINN) was developed by [Raissi et al. \(2017a\)](#), which exploits physics-based regularization in the neural network architecture for solving partial differential equations (PDEs) ([Raissi et al. 2017b, 2019; Raissi, 2018](#)). [Fraces et al. \(2020\)](#) applied PINN framework for solving Buckley-Leverett problems, and it has been extended for uncertainty analysis ([Fraces and Tchelepi, 2021, 2023](#)). The physics-informed discrete learning that employs CNNs in the architecture has been shown to provide better scalability and faster convergence for solving PDEs ([Gao et al., 2021; Ren et al., 2022; Wandel et al., 2022; Zhang, 2022; Bi et al., 2023; Zhang et al., 2023; Zhao et al., 2023](#)). The PINN approaches have been tested on simple CO₂ injection problems, in which they are aiming to simulate CO₂ saturation or concentration distribution ([Shokouhi et al., 2021; Du et al., 2023; Pour and Voskov, 2023; Sheth et al., 2023](#)).

Incorporation of physical laws into the neural network architecture reduces the dependency on the data and mitigates the risk of overfitting. In order to maintain efficiency while enhancing the predictive power and physical interpretability, we utilize reduced physics models in the physical consideration part of the hybrid models. The neural network part of the hybrid models is supposed to learn the complex physics that is omitted in the reduced physics models ([Nagao et al., 2024a](#)).

In the following sections, we first explain the methodology of the proposed hybrid models. Next, we apply the proposed method to a 3D benchmark case to compare the prediction accuracy against an existing pure machine learning model. The estimated reservoir connectivity map by the proposed hybrid model is compared to the streamline flux allocation based on full-physics numerical simulations. To demonstrate the practical feasibility of the proposed framework, we apply it to a CO₂-EOR field case for identification of reservoir connectivity. [Table 1](#) summarizes the advantages and limitations of existing methods for reservoir connectivity identification and production forecasting.

2. Methodology

We propose two different hybrid modeling strategies: physics-based data preprocessing with CRM and a PINN combined with a reduced physics model. These two hybrid model strategies are developed and applied to water flooding cases ([Nagao et al., 2023a; 2024](#)). This paper extends the framework to CO₂ miscible flooding that considers three phase flow system. In this section, we first briefly describe the problem of interest. Next, we review the CRM and pure machine learning models.

Finally, we explain our proposed hybrid models.

2.1. Problem description and assumptions

We are interested in CO₂-EOR reservoirs with large number of producers and injectors. The producers have three phase fluid production. The available data consists of multiphase production rate and CO₂ injection rate. The proposed hybrid machine learning models are developed to rapidly identify correlation between the historical well measurements. If the historical bottom hole pressure (BHP) is also available, it can be incorporated in the hybrid models. The hybrid models are trained solely based on the historical well measurements, and the trained models can be used for two different purposes. First, future production rates can be predicted by the models. Second, trained hybrid models can be used for estimating the reservoir connectivity between wells. Reservoir connectivity is very useful information for optimizing reservoir management to maximize the hydrocarbon production and CO₂ storage capacity. Because the hybrid models are calibrated solely based on historical well measurements, it is not required to generate training datasets by running numerical simulations in our hybrid machine learning framework. However, it is expected that sufficient historical data is available to ensure accuracy and robustness of the hybrid models.

2.2. Capacitance resistance model (CRM)

The Capacitance Resistance Model (CRM) has been widely used in oil and gas industry to understand the performance of water flooding and CO₂-EOR reservoirs. The method utilizes injection rates as input signal and total production rates as the output signal to calibrate a simplified physics-based model with analytical solution ([Al-Yousef, 2006; Yousef et al., 2006; Sayarpour et al., 2007; Sayarpour, 2008](#)). The CRM is calibrated by optimizing several physical parameters, including the well productivity, injection allocation factor, and a response delay constant. The allocation factor represents the connectivity and fluid flux between each injector-producer pair. The CRM is commonly used because of its efficiency, capability of future production forecasting, and reservoir connectivity identification. For a single injector and producer case, the material balance equation can be written as follows:

$$c_t V_p \frac{d\bar{P}}{dt} = i(t) - q(t) \quad (1)$$

where c_t is the total compressibility, V_p is the reservoir pore volume, $\bar{P}$ is the average reservoir pressure, $i(t)$ is injection rate, and $q(t)$ is total production rate. The total production rate can be approximated by the following equation:

$$q(t) = J(\bar{P} - P_{wf}) \quad (2)$$

Where J is the productivity index, P_{wf} is the BHP at the producer. From Eqs. (1) and (2), the average reservoir pressure can be eliminated, and it gives the first-order ordinary differential equation for the CRM as ([Al-Yousef, 2006](#)):

$$\tau \frac{dq}{dt} + q(t) = i(t) - \tau J \frac{dP_{wf}}{dt} \quad (3)$$

where $\tau = c_t V_p / J$ is a time constant. The analytical solution for $q(t)$ can be written as ([Al-Yousef, 2006](#)):

$$q(t) = q(t_0) e^{-\frac{t-t_0}{\tau}} + \frac{1}{\tau} e^{-\frac{t}{\tau}} \int_{t_0}^t e^{\xi/\tau} i(\xi) d\xi - J \left(P_{wf}(t) - e^{-\frac{t-t_0}{\tau}} P_{wf}(t_0) \right) + \frac{J}{\tau} e^{-\frac{t}{\tau}} \int_{t_0}^t e^{\xi/\tau} P_{wf}(\xi) d\xi \quad (4)$$

where t_0 is the initial time at which $q(t) = q(t_0)$ and ξ is a variable of

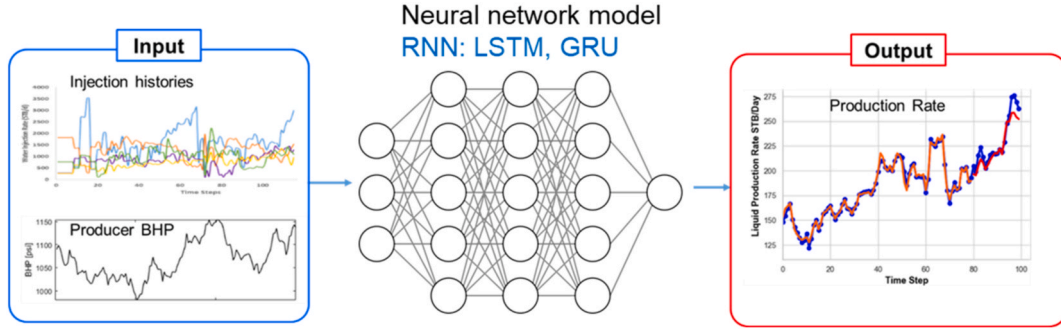


Fig. 1. Schematic of pure machine learning model using Recurrent Neural.

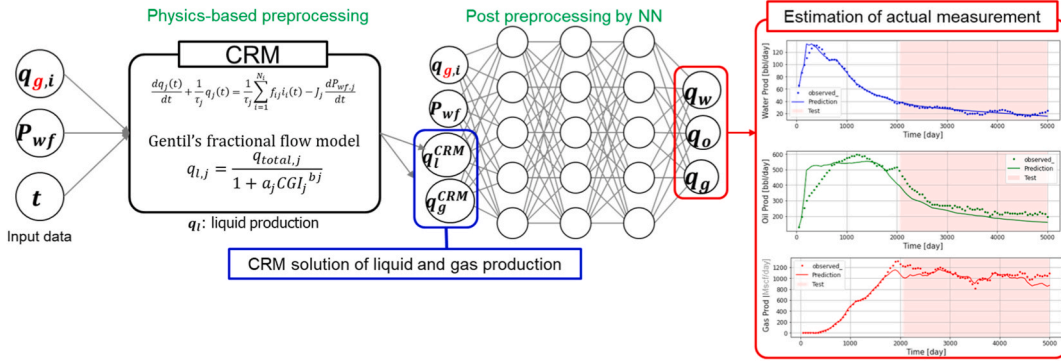


Fig. 2. Framework of the physics-based data preprocessing with CRM for three phase production forecasting.

integration. If the historical BHP data is not available, we assume a constant value over time.

For the multiple injector and producer case, Sayarpour (2008) developed two different types of extension of the above expression, CRMP and CRMIP. The CRMP has one time constant for each producer, and the CRMIP has one time constant for each injector-producer pair. The CRMP is used in the applications of CRM in this study. The governing equation of the CRMP is written as:

$$\tau_j \frac{dq_j(t)}{dt} + q_j(t) = \sum_{i=1}^{n_i} f_{ij} i_i(t) - \tau_j J_j \frac{dp_{wf,j}}{dt} \quad (5)$$

where n_i is the number of injectors, f_{ij} is the fraction of injection rate of injector i contributing to the production rate at producer j . In this model f_{ij} , τ_j and J_j are the calibration parameters to fit the total production rates.

For estimating the liquid production and gas production rates during CO₂ injection, an empirical power law fractional flow model proposed by Gentil (2005) is used in our applications. The liquid production rate is given as follows:

$$q_{\text{liquid},jk} = \frac{q_{\text{total},jk}}{1 + a_j C G I_{jk}^{b_j}} \quad (6)$$

where $q_{\text{total},jk}$ is the estimated total production rate of producer j at timestep k , $C G I_{jk}$ is the cumulative CO₂ injected over all time periods until timestep k contributing to producer j , which is given by:

$$C G I_{jk} = \sum_{k=1}^{n_t} \sum_{i=1}^{n_i} f_{ij} i_{ik} \quad \text{for all } j \text{ and } k \quad (7)$$

where n_t is number of timesteps and i_{ik} is CO₂ injection rate of injector i at timestep k . a_j and b_j in Eq. (6) are fractional flow model parameters for each producer j . These parameters can be estimated by linear regression with simple manipulation (Weber et al., 2009). We use historical liquid

production rate to find these parameters, and the calibrated parameters can be used to predict future liquid and gas production rate (Sayarpour, 2008; Nguyen, 2012; Tao and Bryant, 2015).

2.3. Pure machine learning models for production forecasting

Pure machine learning models have been used to find statistical correlation between injection rate, production rates, and pressure data. Once the machine learning model is trained, it can be used for identifying reservoir connectivity using the concept of variable importance (Chen et al., 2021). Fig. 1 shows the schematic of the pure machine learning workflow for production forecasting. Recurrent Neural Networks (RNN) is commonly used for time series data, including Long Short-term Memory (LSTM) and Gated Recurrent Unit (GRU). The input of the models includes injection rate at injectors impacting the producer and the producer's BHP. A search radius from the producer determines the injection wells that are included in the input data for production forecasting, and the radius needs to be pre-determined before starting the neural network training. The output of the models is multiphase production rate. The implementation of the pure machine learning framework is quite simple and quick. In the following comparison studies, LSTM is used as the pure machine learning model.

2.4. Hybrid model 1: physics-based data preprocessing

Fig. 2 describes the architecture of the physics-based data preprocessing, and Fig. 3 shows the detailed workflow. The data preprocessing is composed of two steps. First, a reduced physics model is used to obtain an approximate physics-based solution efficiently, which we call physics-based preprocessing. Next, the approximate physical solution serves as the input to the neural network model. The machine learning model estimates the multiphase production rate based on the physically augmented input features. For the preprocessing part in our applications, we use the CRM with Gentil's fractional flow model (Gentil,

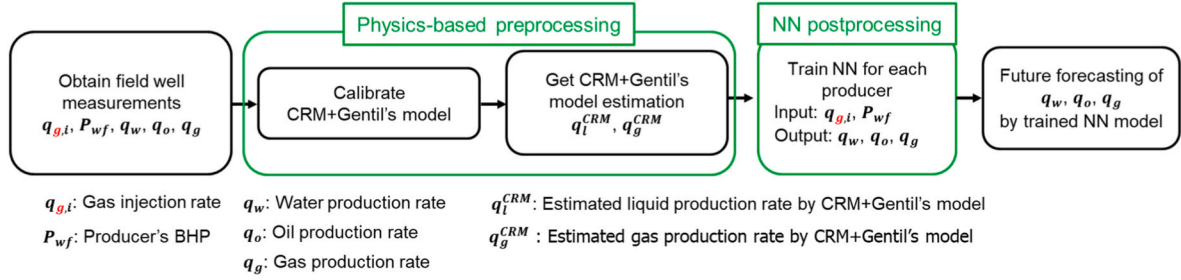


Fig. 3. Flowchart of physics-based data preprocessing method.

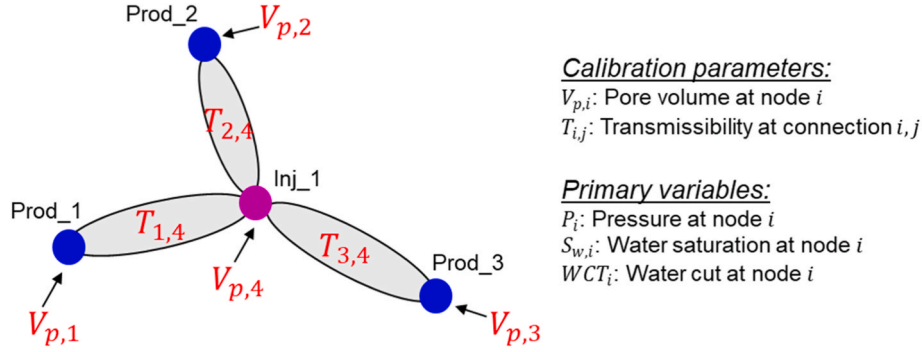


Fig. 4. — Schematic of the reduced physics model for a simple example.

2005). The CRM takes CO₂ injection rate, producer's BHP, and time as input data, and it estimates the liquid and gas production rates. Due to the simplified physics assumption in CRM, it could introduce some estimation error. The neural network part attempts to further improve the predictive performance given from CRM.

The feed-forward neural network model is built for each producer as the data-driven part of this hybrid model. The machine learning part (neural network model) is designed to learn the complex physics such as dynamic allocation factor and transient PI, enhancing the production forecasting performance.

2.5. Hybrid model 2: physics-informed neural network (PINN) model

In this section, we develop a PINN framework for our application. The PINN incorporates a physics-based regularization term into the loss function of the neural network model as follows:

$$Loss = L_{data_misfit} + L_{phy_regularization} \quad (8)$$

where the training loss L_{data_misfit} represents standard data misfit error (e. g. RMSE, MAE, MAPE, etc.) between the model estimation and actual measurements, $L_{phy_regularization}$ is physics-based regularization term to prevent that trained neural network model deviates from the fundamental physical laws, for example material balance, momentum balance, and energy balance, etc. The physics-based regularization term will be minimized during the neural network training along with the data misfit term. In our applications, we use the residual of the governing material balance equations for the physics-based regularization term $L_{phy_regularization}$.

2.5.1. Reduced physics model

In order to maintain the computational efficiency, we use a reduced physics model for the physics-based residual term. Fig. 4 shows the schematic of the reduced physics model that is used in our PINN formulation. This reduced physics model is inspired by INSIM (Zhao et al., 2016). In this example, we have three producers and one injector at the center. We first define a node for each well. Next, we make

connections between producer and injector nodes. Each connection has a transmissibility, and each well node has a pore volume as calibration parameters.

For the three-phase oil, water, and gas black oil type system, we can now write the governing equations as follows:

$$\sum_{j=1}^n T_{w,i,j} [P_j - P_i] + q_{w,i} - \frac{\partial}{\partial t} \left(\frac{V_{p,i} S_{w,i}}{B_{w,i}} \right) = 0 \quad (9)$$

$$\sum_{j=1}^n T_{o,i,j} [P_j - P_i] + q_{o,i} - \frac{\partial}{\partial t} \left(\frac{V_{p,i} S_{o,i}}{B_{o,i}} \right) = 0 \quad (10)$$

$$\sum_{j=1}^n T_{g,i,j} [P_j - P_i] + q_{g,i} - \frac{\partial}{\partial t} \left(\frac{V_{p,i} S_{g,i}}{B_{g,i}} \right) = 0 \quad (11)$$

$$S_{w,i} + S_{o,i} + S_{g,i} = 1.0 \quad (12)$$

where we have transmissibility of water, oil, and gas phase for each connection (i, j) $T_{w,i,j}$, $T_{o,i,j}$, $T_{g,i,j}$, pore volume associated with each well node $V_{p,i}$, formation volume factor of water, oil, and gas phase $B_{w,i}$, $B_{o,i}$, $B_{g,i}$, sink/source terms $q_{w,i}$, $q_{o,i}$, $q_{g,i}$, reservoir pressure at each pore volume associated with a well node P_i , saturation of each phase at each pore volume associated with a well node $S_{w,i}$, $S_{o,i}$, $S_{g,i}$, and n is the number of regions adjacent to the region i . Eq. (9) is water phase material balance equation, Eq. (10) is oil phase material balance equation, and Eq. (11) is gas phase material balance. In this study, it is assumed that the solution gas is negligible. In these governing equations, the primary variables include the pressure P_i , water saturation $S_{w,i}$, and water cut at each pore volume. The transmissibility, pore volume and multiphase production rate can be calculated as follows:

$$T_{a,i,j} = \left(\frac{kA}{L} \right) \frac{k_{ra}}{\mu_a B_a} = T_{ij} \frac{k_{ra}}{\mu_a B_a} \quad (13)$$

$$V_{p,i} = V_{p,i}^0 [1 + c_r (P_i - P^0)] \quad (14)$$

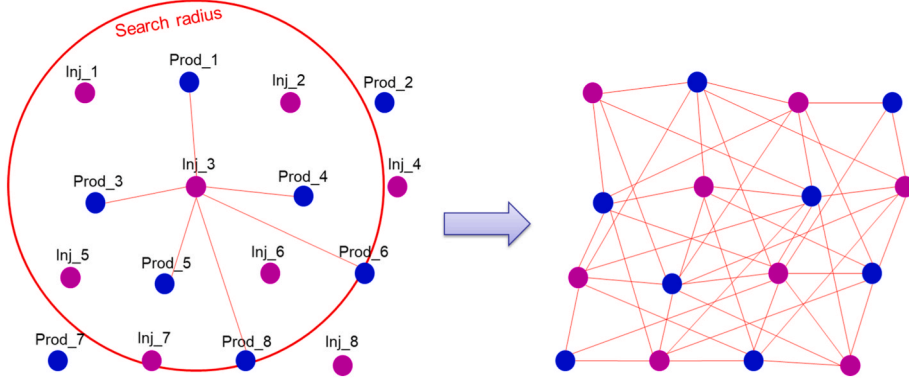


Fig. 5. Construct reservoir network system using search radius.

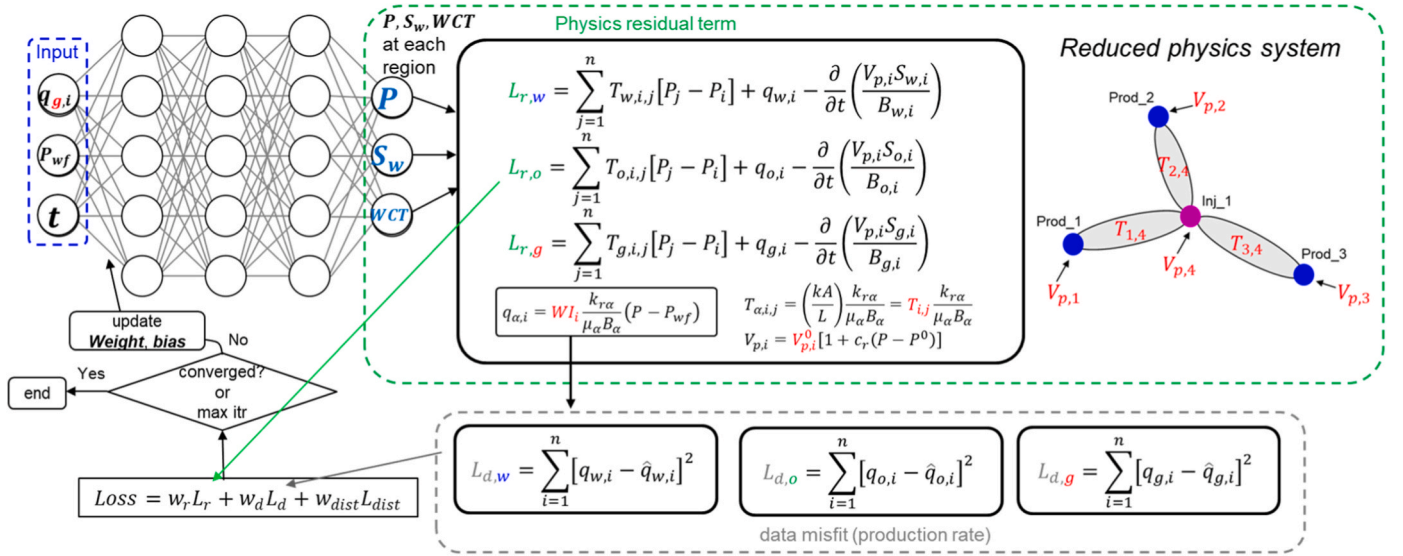


Fig. 6. PINN architecture combined with reduced physics model.

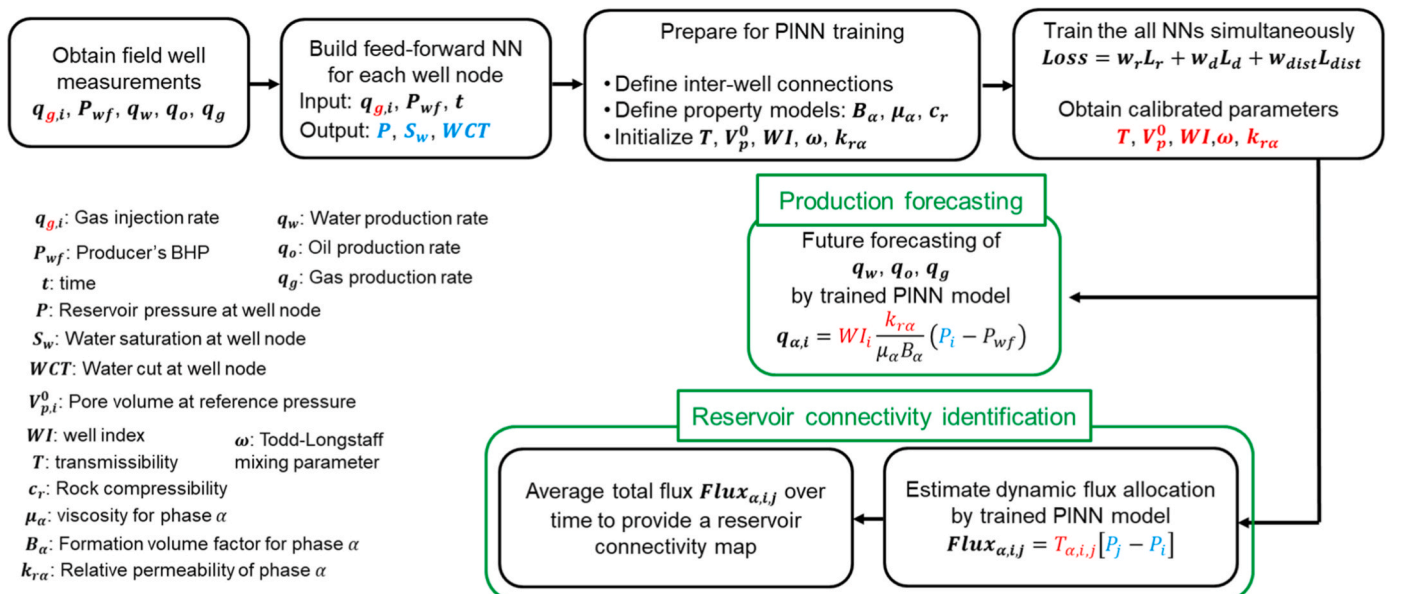


Fig. 7. Flowchart of the PINN method.

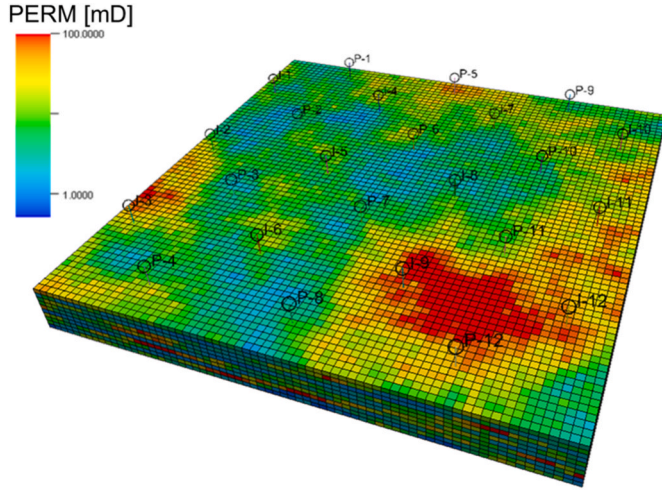


Fig. 8. Synthetic reservoir description (permeability map).

Table 2
Synthetic case reservoir property and initial conditions.

Property name	Value
Porosity	0.25
Depth of the top layer	8861 ft
Initial water saturation	0.5
Initial gas saturation	0.0
Initial reservoir pressure	1000 [psi]

$$q_{\alpha,i} = \mathbf{W} \mathbf{I}_i \frac{k_{ra}}{\mu_{\alpha} B_{\alpha}} (P_i - P_{wf}) \quad (15)$$

where we have absolute permeability k , connection length L , connection area A , pore volume at reference condition $V_{p,i}^0$, rock compressibility c_r , phase index α , well index at producer node i , $\mathbf{W} \mathbf{I}_i$, and bottomhole pressure P_{wf} . In history matching process, the bolded parameters in Eq. (13)–(15) could be treated as calibration parameters to match well response data. To account for the miscibility of injected gas into oil phase, we use the Todd-Longstaff model (Todd and Longstaff, 1972).

The Todd-Longstaff model is widely used to model miscible displacement during gas injection in oil reservoirs. The model introduces an empirical parameter, which lies between 0 and 1 to describe the level of dispersion. When the Todd-Longstaff mixing parameter is 1, it means that the hydrocarbon components are fully mixed with the injected fluid. When it is 0, it is considered as immiscible displacement. The effective oil and gas viscosities are given by:

$$\mu_{o,eff} = \mu_o^{1-\omega} \mu_m^{\omega} \quad (16)$$

$$\mu_{g,eff} = \mu_g^{1-\omega} \mu_m^{\omega} \quad (17)$$

where ω is the Todd-Longstaff mixing parameter lies between 0 and 1, and μ_m is oil and gas mixture viscosity. Details of the mixture viscosity, density, and three-phase relative permeability calculations are described in Appendix A. Although this single-tank formulation of reduced physics model could possibly omit some complex physics, such as facies contrast or vertical compartmentalization in heterogeneous reservoirs, our proposed hybrid model aims to learn the missing complex physics by the machine learning part of the algorithm in the PINN architecture. In that way, we can maintain both accuracy and computational efficiency.

2.5.2. Preprocessing for the reduced physics model

In the proposed reduced physics-based model, the connections between wells need to be determined before PINN training. In our framework, we determine the well connections based on the distance and preset search radius. We first specify the search radius, where same search radius is used for all wells. Fig. 5 shows the schematic of reservoir network construction. In this example, the search radius is applied to “Inj_3”, and we connect all producers to “Inj_3” within the specified search radius. By applying this process for all injectors, we can create reservoir network system as shown in Fig. 5.

2.5.3. PINN architecture combined with reduced physics model

Fig. 6 shows the architecture of the PINN combined with the reduced physics model. As input data of the PINN architecture, we have used CO₂ injection rate, BHP of producers, and time information. If the historical BHP data is not available, we could assume constant BHP over time. The feed-forward neural network computes three primary variables, pressure, water saturation, and water cut at each pore volume associated with a well node for all time steps from the input data.

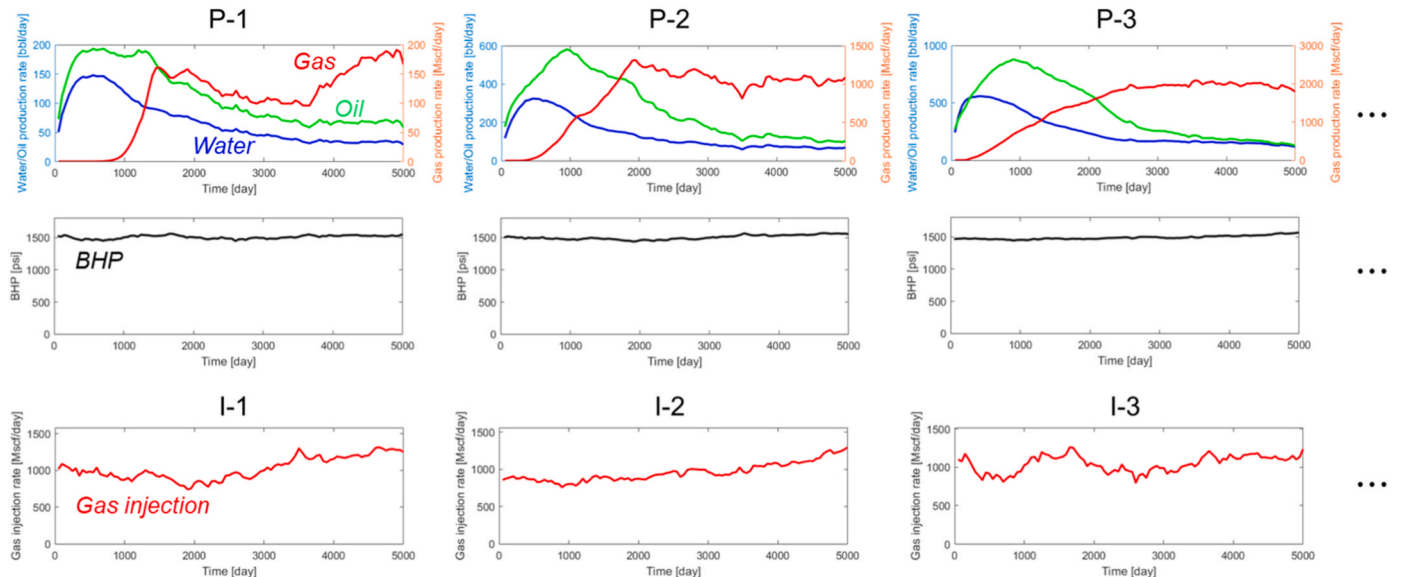


Fig. 9. Production, injection, and pressure measurements of selected wells for the synthetic case.

Table 3
Hyper parameters for LSTM.

Parameter	Value
Search radius [ft]	1061
Number of hidden layers	2
Maximum number of epochs	300
Learning rate	0.001
Train, validation, and test ratio	0.35, 0.05, 0.6
Train, validation, and test ratio [days]	1750, 250, 3000

Table 4
LSTM architecture.

Hidden layer name	Number of nodes	Activation function
LSTM	16	ReLU
Dense	32	ReLU

Table 5
Hyper parameters for GRU.

Parameter	Value
Search radius [ft]	1061
Number of hidden layers	2
Maximum number of epochs	300
Learning rate	0.001
Train, validation, and test ratio	0.35, 0.05, 0.6
Train, validation, and test ratio [days]	1750, 250, 3000

2.5.4. PINN loss function

We use the reduced physics model explained above for the calculation of the physics-based regularization term in the PINN loss function. Based on the computed primary variables from the feed-forward neural network, we calculate the residual of three phase material balance equations for each pore volume associated with a well node, $L_{r,w}$, $L_{r,o}$, and $L_{r,g}$.

$$L_{r,w} = \sum_{j=1}^n T_{w,ij} [P_j - P_i] + q_{w,i} - \frac{\partial}{\partial t} \left(\frac{V_{p,i} S_{w,i}}{B_{w,i}} \right) \quad (18)$$

$$L_{r,o} = \sum_{j=1}^n T_{o,ij} [P_j - P_i] + q_{o,i} - \frac{\partial}{\partial t} \left(\frac{V_{p,i} S_{o,i}}{B_{o,i}} \right) \quad (19)$$

$$L_{r,g} = \sum_{j=1}^n T_{g,ij} [P_j - P_i] + q_{g,i} - \frac{\partial}{\partial t} \left(\frac{V_{p,i} S_{g,i}}{B_{g,i}} \right) \quad (20)$$

These residual terms are included in the loss function. The time derivative in the accumulation terms is calculated by numerical derivative, as it showed more stable training performance compared to automatic differentiation in our study. The loss function also has the data misfit terms for three phase production rates. We calculate the multiphase production rate using Eq. (15), and the data misfit is calculated as the square error between the actual measurement and estimated production rates:

$$L_{d,\alpha} = \sum_{i=1}^{n_p} [q_{\alpha,i} - \hat{q}_{\alpha,i}]^2 \quad (21)$$

where we have actual measurement of multiphase production rate $q_{\alpha,i}$, estimated multiphase production rate $\hat{q}_{\alpha,i}$, phase index α , and number of producers n_p . Now the loss function is the summation of data misfit and

Table 6
GRU architecture.

Hidden layer name	Number of nodes	Activation function
GRU	16	ReLU
Dense	32	ReLU

Table 7
Hyper parameters for physics-based data preprocess method.

Parameter	Value
Search radius [ft]	1061
Number of hidden layers	2
Maximum number of epochs	300
Learning rate	0.001
Train, validation, and test ratio	0.35, 0.05, 0.6
Train, validation, and test ratio [days]	1750, 250, 3000

Table 8
Neural network architecture for physics-based data preprocess method.

Hidden layer name	Number of nodes	Activation function
LSTM	16	ReLU
Dense	32	ReLU

Table 9
Hyper parameters for PINN.

Parameter	Value
Search radius [ft]	1678
Number of hidden layers	2
Maximum number of epochs	25000
Learning rate	0.001
Train, validation, and test ratio	0.35, 0.05, 0.6
Train, validation, and test ratio [days]	1750, 250, 3000

physics-based residual terms, which can be written as follows:

$$Loss = w_r L_r + w_d L_d \quad (22)$$

$$L_r = w_w L_{r,w} + w_o L_{r,o} + w_g L_{r,g} \quad (23)$$

$$L_d = L_{d,w} + L_{d,o} + L_{d,g} \quad (24)$$

where $L_{r,w}$, $L_{r,o}$, and $L_{r,g}$ are residual terms of three phase material balance equations, $L_{d,w}$, $L_{d,o}$, and $L_{d,g}$ are the data misfit terms of three phase production rates, w_w , w_o , and w_g are the weights associated with three phase residual terms, and w_r and w_d are weights for residual loss and data misfit loss. These weights are introduced in the calculation of loss function to control the effect of the physical term. In a previous study, the selection of optimal weight values is addressed (Bischof and Kraus, 2025).

2.5.5. Distance-based penalization in PINN training

In the PINN training formulation above, the information of distance between wells is not included. Generally, longer distance connections tend to have smaller flux allocations. In order to properly represent this, we introduce the concept of distance-based penalty function. This idea is originally proposed by Chen et al. (2021) and Sen et al. (2022) and modified to adapt to our PINN framework. In the loss function, additional distance-based regularization term is added as follows:

$$Loss = w_r L_r + w_d L_d + w_{dist} L_{dist} \quad (25)$$

$$L_{dist} = \sum_{i_c=1}^{N_c} T_{i_c} Dist_{i_c} \quad (26)$$

where T_{i_c} and $Dist_{i_c}$ are the transmissibility and distance at connection i_c , and N_c is number of connections. During the PINN training, this loss

Table 10
Neural network architecture for PINN.

Hidden layer name	Number of nodes	Activation function
Dense	60	ReLU
Dense	60	ReLU

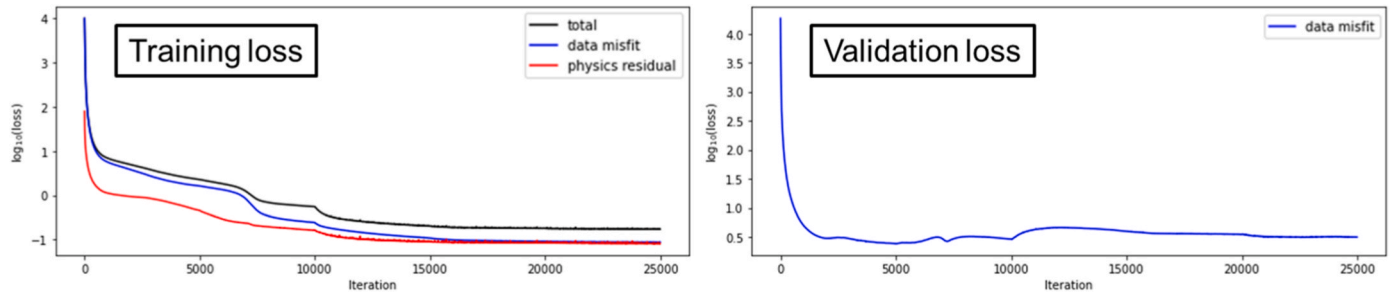


Fig. 10. Training and validation data loss trend during the PINN training for the synthetic case.

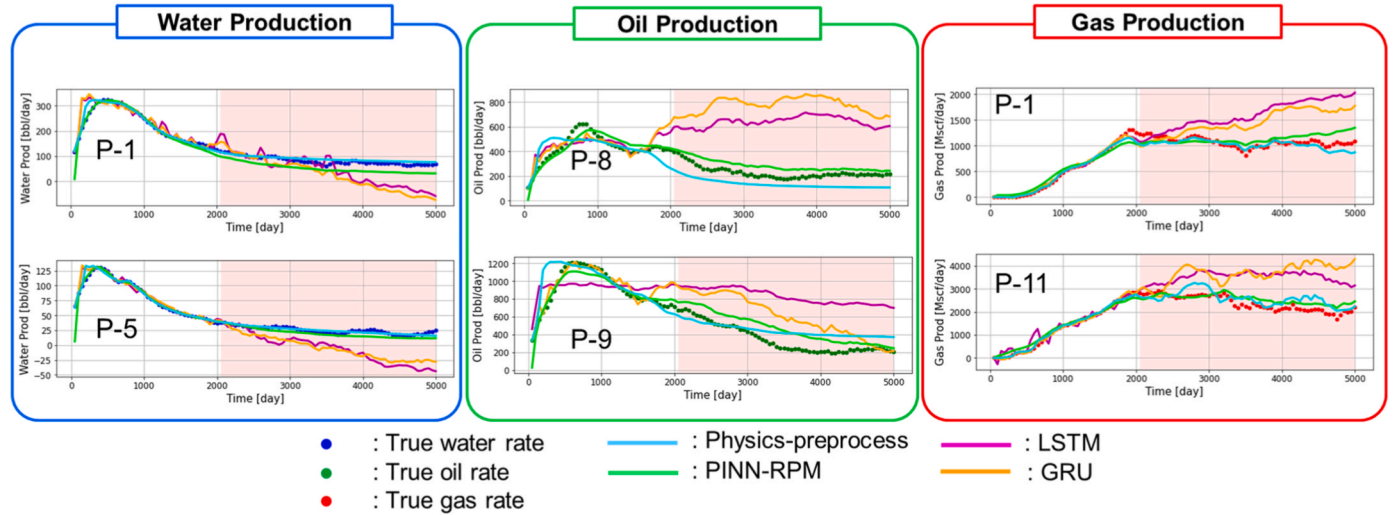


Fig. 11. Prediction results of multiphase production rates of selected two wells for the synthetic case.

function in Eq. (25) is minimized instead of Eq. (22) to calibrate the model. By adding L_{dist} term, the transmissibility at long distance connection becomes small so that long distance connections are automatically penalized more. Penalty coefficient w_{dist} determines the magnitude of the distance-based penalization, which is a tuning parameter of the framework. To find the optimal value of the penalty coefficient, we run PINN training using multiple different w_{dist} values. Generally, larger penalty coefficients lead to larger estimation errors in training data because the penalty term forces to reduce the number of reservoir connections and simplify the model. The optimal coefficient value can be selected by identifying the point where the training estimation error starts to increase, as you gradually increase the penalty coefficient w_{dist} . The detailed steps to find the optimal penalty coefficient

are demonstrated in Appendix B.

2.5.6. PINN training

Neural network training calibrates the neural network parameters, weight and bias of each neuron, as well as physical parameters used in the residual calculation in order to minimize the loss function. The physical calibration parameter includes pore volume of each well node, transmissibility for each connection, well index at each producer node, parameters for relative permeability models, and Todd-Longstaff mixing parameters. The first three parameters are represented by bolded characters in Eq. (13)–(15). Details of relative permeability and Todd-Longstaff mixing parameters are provided in Appendix A. Since the PINN architecture aims to minimize the physics-based residual terms as well as the data misfit, the trained PINN architecture should satisfy the three phase material balance equations within specified tolerance. Also,

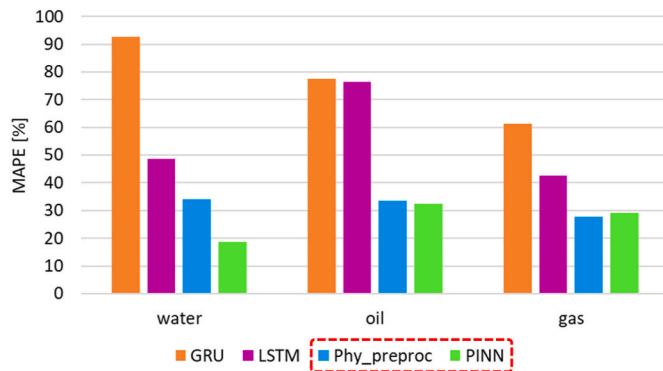


Fig. 12. Comparison of prediction performance between three methods. The error is calculated as mean absolute percentage error of multiphase production rate forecasting (blind-test) for all 22 producers.

Table 11
RMSE of multiphase production forecasting.

	LSTM	GRU	Physics preprocess	PINN
water [STB/d]	48.75	93.04	34.30	18.79
oil [STB/d]	192.53	195.54	84.12	81.90
gas [Mscf/d]	406.29	584.69	263.00	277.36

Table 12
R2 score of multiphase production forecasting.

	LSTM	GRU	Physics preprocess	PINN
water	−2.13	−11.05	−0.78	0.57
oil	−3.64	−4.17	0.12	0.24
gas	0.55	−0.06	0.82	0.69

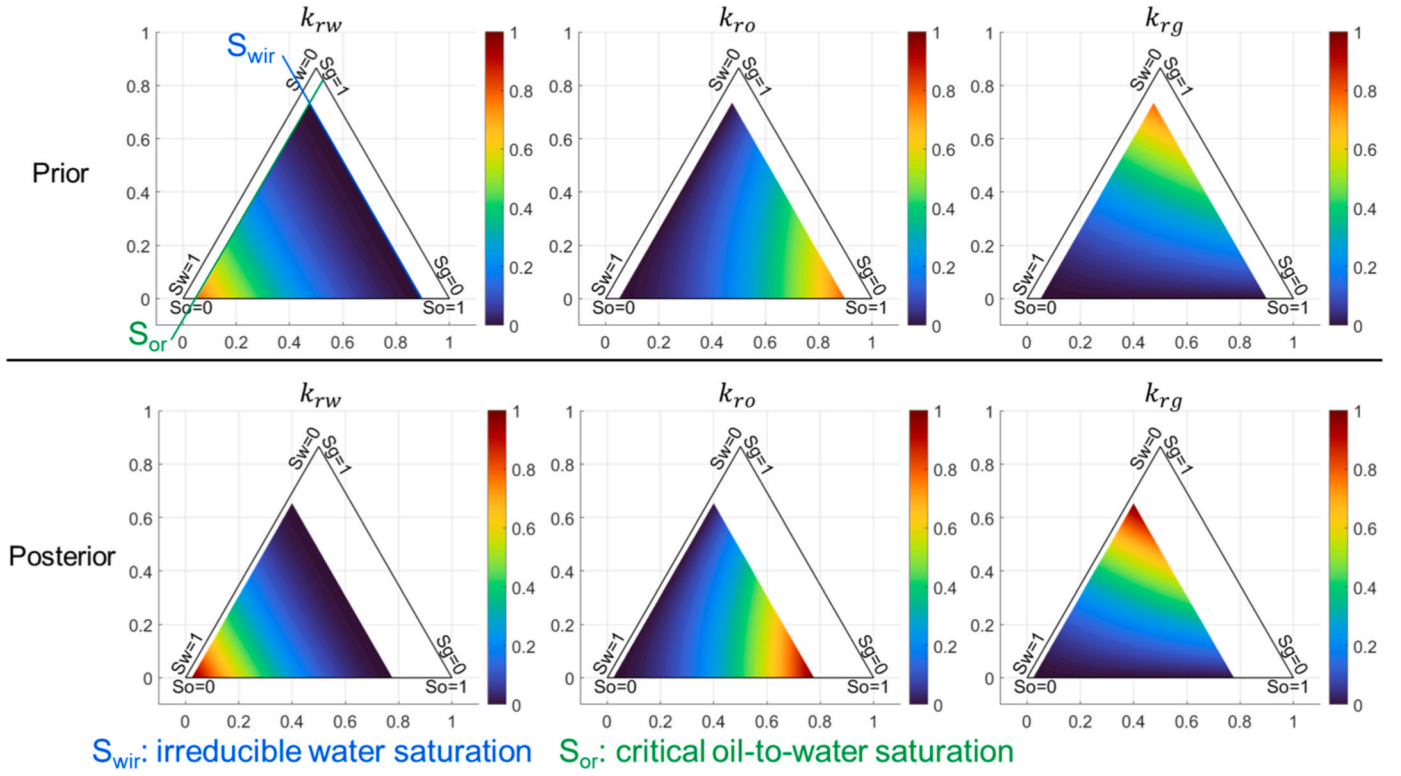


Fig. 13. Three-phase relative permeability models for prior and posterior models.

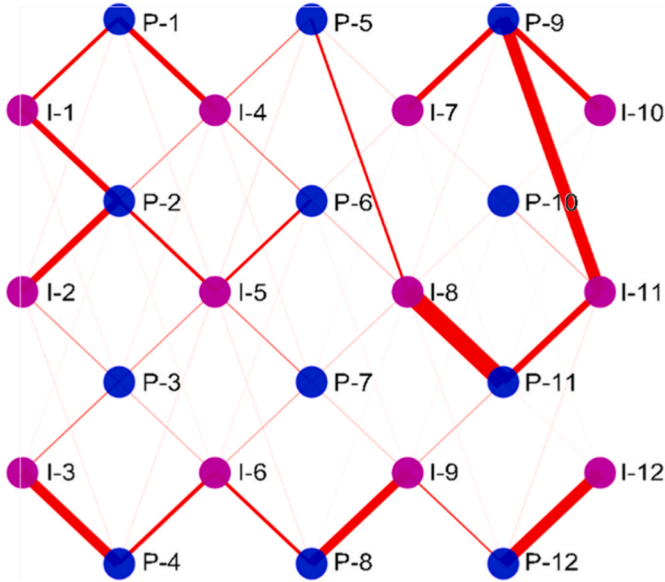


Fig. 14. Reservoir connectivity map from PINN with reduced physics model for the synthetic case.

we get calibrated physical parameters explained above, because those variables are calibrated during the PINN training. Moreover, we obtain dynamic allocation factors after the PINN training. In the proposed reduced physics system, the multiphase flux is calculated for all time steps, which is essentially a dynamic allocation factor. By averaging the dynamic allocation factor for all timesteps, we can obtain the reservoir connectivity. Consideration of the dynamic changes of the allocation factor can enhance the prediction performance. Since the physics-based calculation part is constructed together with the feed-forward neural

network as a single architecture, the backpropagation starting from the loss function goes through the physics-based calculation part, and it propagates to the feed-forward neural network part. Therefore, the gradient with respect to the physical calibration parameters and neural network weight and bias are efficiently calculated with automatic differentiation in the backpropagation process. Fig. 7 shows the detailed flowchart of our proposed PINN method.

3. Validation

In this section, we apply the proposed hybrid models to a synthetic reservoir case. The prediction performance of the proposed hybrid models is compared to pure machine learning to analyze the prediction performance between different methodologies. The following four different approaches are applied in this study.

- LSTM (pure machine learning model)
- GRU (pure machine learning model)
- Physics-based data preprocessing with CRM
- Physics-informed Neural Network (PINN) with a reduced physics model

In the following applications, the deep learning framework is implemented using TensorFlow library in python (Martín et al., 2015). The optimization procedure in the CRM is implemented using SciPy library in python. All calculation was implemented with an Intel(R) Xeon (R) CPU E5-2630 v4 @ 2.20 GHz using a single core.

3.1. Model description

The reservoir model has 69x69x10 grids and the domain size is 5175.0x5175.0x100.0 [ft]. The porosity is uniformly distributed with value of 0.25, and we have a heterogeneous permeability field as shown in Fig. 8. The depth of the top layer is 8861 [ft]. Total 12 producers and 12 injectors are completed in the reservoir, where we have continuous

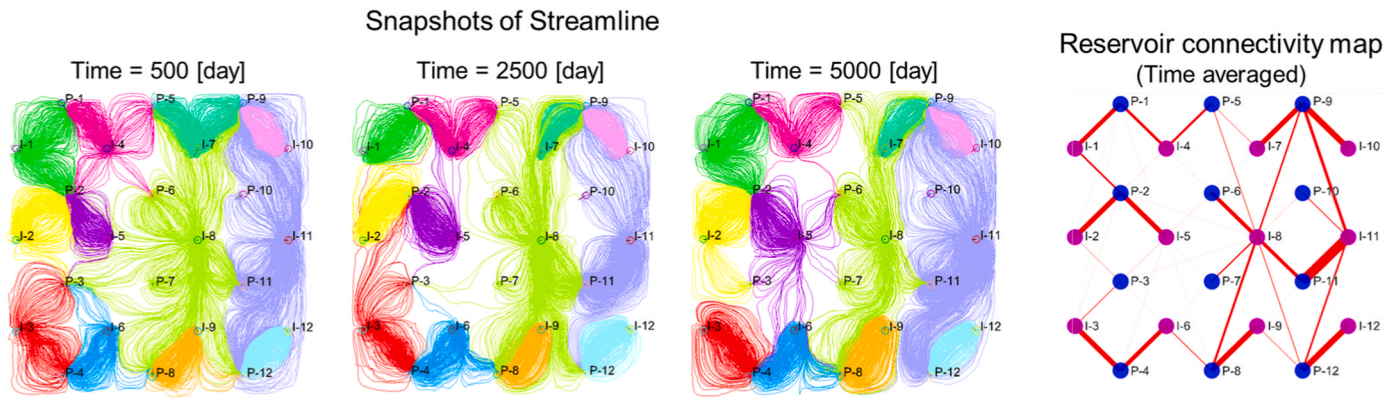


Fig. 15. Streamline tracing map (at selected timesteps) and reservoir connectivity map based on the time averaged flux allocation by streamline for the synthetic case.

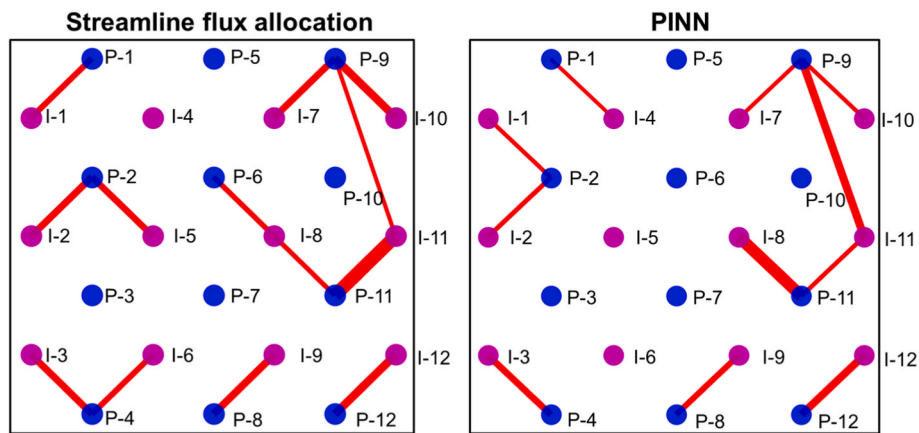


Fig. 16. Comparison of reservoir connectivity maps between streamline-based method and PINN for the synthetic case. Only strong connectivity lines (over 500 [bbl/day]) are shown.

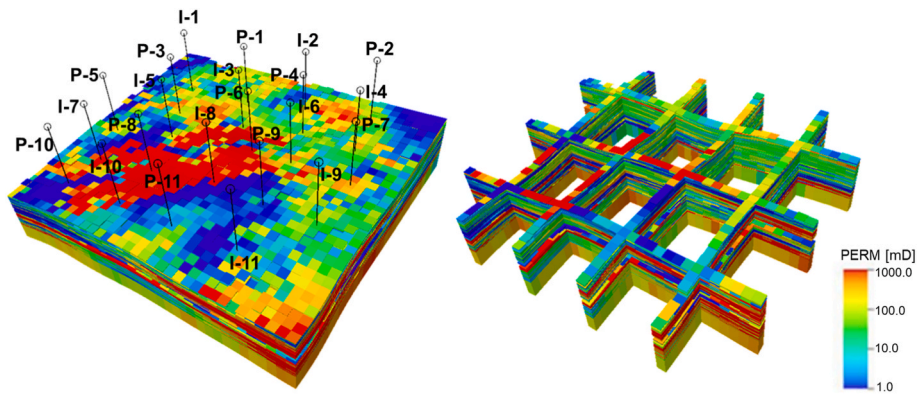


Fig. 17. Permeability distribution of the field case.

CO₂ injection for about 14 years. For initial conditions, water saturation is set at 0.5, gas saturation is 0.0, and initial reservoir pressure is 1000 [psi]. Reservoir properties and initial conditions are summarized in Table 2. Commercial simulator (Schlumberger, 2022) is used to run flow simulation and create the benchmark well histories. The all producers have variable BHP constraint, and all injectors have rate constraint in the simulation. The input data of the machine learning and hybrid models is composed of the CO₂ injection rate of 12 injectors and BHP at 12 producers. Total 2000 days of historical well measurements are available from all the wells, and we conduct future prediction of

multiphase production rate for 3000 days. The well measurements for some sample wells are shown in Fig. 9.

3.2. Model training

3.2.1. LSTM and GRU

We estimate the three phase production rates based on the input data, including time step information, BHP of the producer, and water injection rate of the surrounding injectors. For neural network training purposes, the last 250 days of the historical well measurements are

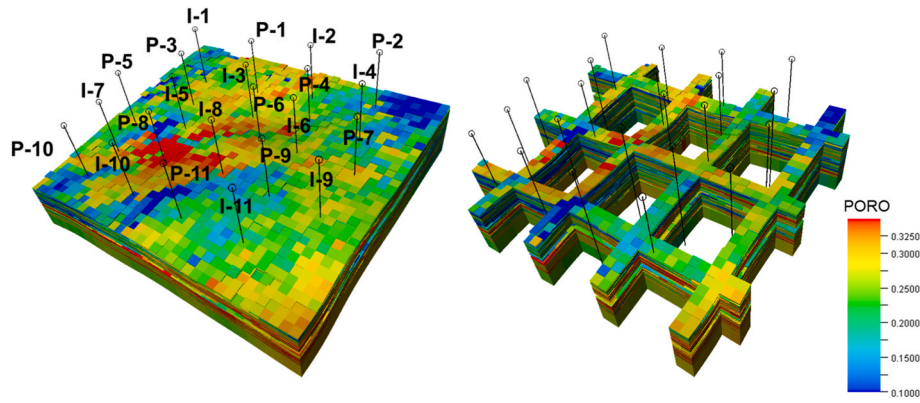


Fig. 18. Porosity distribution of the field case.

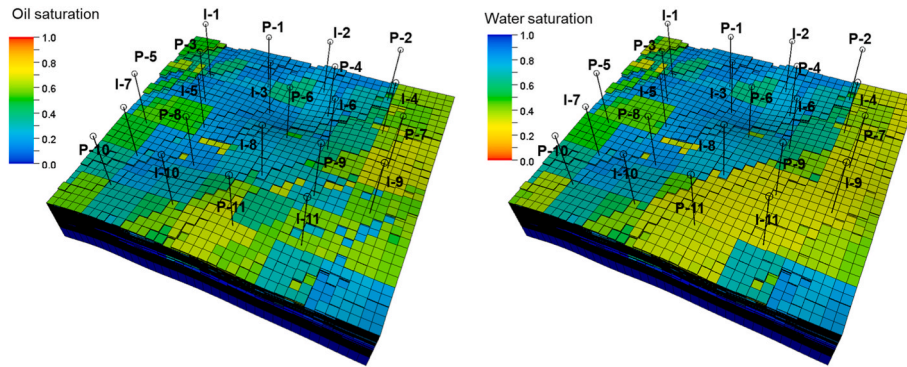


Fig. 19. Initial oil and water saturation for the field case.

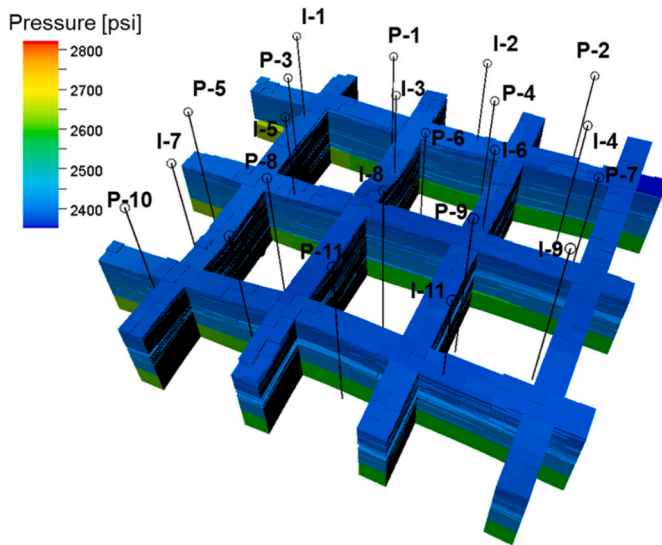


Fig. 20. Initial pressure condition of the field case.

defined as validation dataset. By monitoring the trend of the validation data loss function with respect to the training iteration, we can stop training and prevent overfitting. The LSTM and GRU models are built and trained for each producer. The injectors within a predefined search radius from the producer are included in the input data. Several different neural network architectures were tested by changing the number of layers and nodes in order to find the optimal size of the architecture. The hyperparameters and neural network architecture are described in

Table 3, Table 4, Tables 5, and Table 6. Dense is a typical type of neural network node used to build fully connected neural network architectures.

3.2.2. Physics-based data preprocessing with CRM

We obtain approximate physics solution by using CRM, and then neural network is used to improve the prediction accuracy of the CRM. The neural network models are built and trained for each producer same as in the LSTM. The input data is composed of the estimated liquid and gas production rate provided by CRM, BHP of producer, and gas injection rates within the predefined search radius. The hyperparameters and neural network architecture are described in Tables 7 and 8.

3.2.3. PINN

First, the reduced physics system is built by connecting the producers and injectors based on search radius. The feed-forward neural network is constructed for each producer and all of them are trained simultaneously including the physical part of the PINN architecture. The hyperparameters and neural network architecture are described in Tables 9 and 10. Fig. 10 shows that the trend of the training and validation data loss decrease during the PINN training. For training data loss, three solid lines are plotted, including data misfit term, physics-based residual term, and total loss function. For validation data loss, data misfit term is plotted to check overfitting behavior. Both training loss and validation loss decrease with training iteration and stay flat at the end, which indicates that the PINN training is properly conducted without overfitting.

3.3. Prediction result

Fig. 11 shows the prediction result of three phase production rates for selected two wells in the reservoir. The pink window represents the blind test part, and the models are calibrated using the well

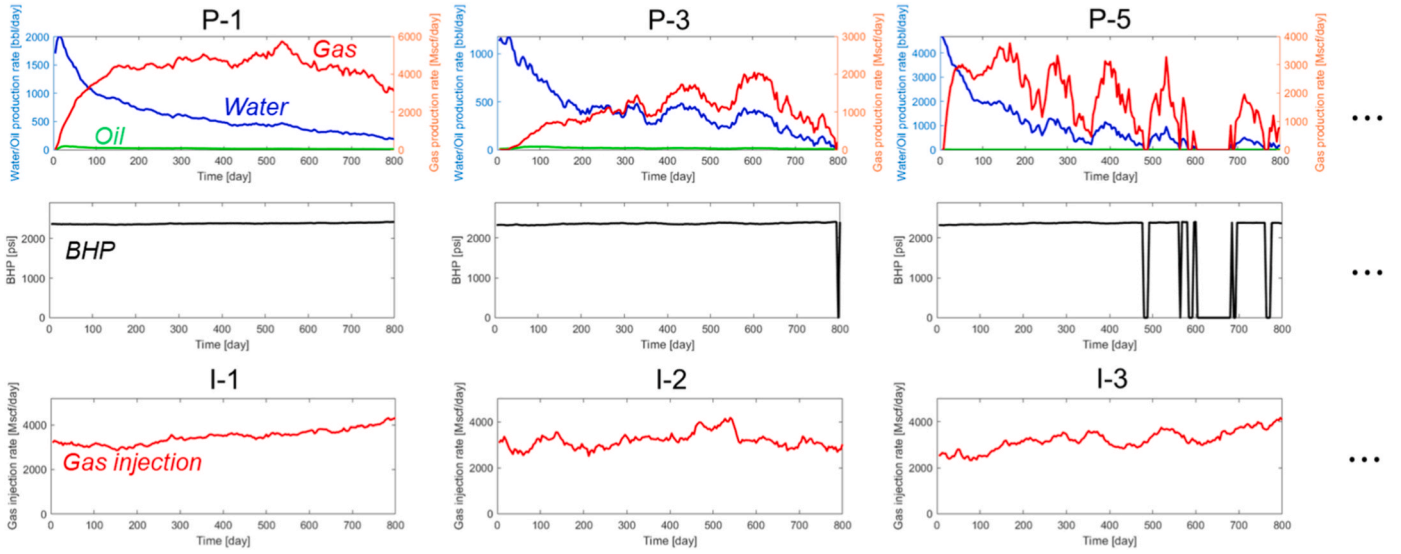


Fig. 21. Production, injection, and pressure measurements of selected wells for the field case.

Table 13

Hyper parameters of PINN for field application.

Parameter	Value
Search radius [ft]	1492
Number of hidden layers	2
Maximum number of epochs	20000
Learning rate	0.001
Train, validation, and test ratio	0.75, 0.05, 0.2
Train, validation, and test ratio [days]	600, 40, 160

Table 14

Neural network architecture of PINN for field application.

Hidden layer name	Number of nodes	Activation function
Dense	32	ReLU
Dense	32	ReLU

measurements within the white window in the plots. The LSTM and GRU models (pure machine learning) tends to overfit the training data and does not provide reasonable prediction accuracy. Since we conducted long term prediction in this case, the trend of production rate can be significantly different between historical production data and the prediction part. In such cases, pure machine learning models tend to suffer because they are solely based on the statistical data correlation. On the other hand, the proposed two physics-informed machine learning models provided reasonable prediction accuracy. By incorporating physics into the architecture, the long-term trend of three phase production rates is successfully estimated. The future prediction

performance (blind test) of multiphase production rate for all 12 producers are summarized in Fig. 12, where the averaged prediction error of 12 producers is calculated as mean absolute percent error (MAPE). It indicates that the proposed two physics-informed machine learning models provide better predictive power over LSTM for this application. Two other error metrics, root mean square error (RMSE) and R2 score, are summarized in Tables 11 and 12.

The relative permeability model is calibrated through PINN training, and the prior and posterior models are described in Fig. 13. Details of the relative permeability model can be found in Appendix A. After PINN training, the end point relative permeability of both water phase and hydrocarbon phase increased, and critical oil-to-water saturation is increased. As a result, mobility of hydrocarbon phase increased after model calibration by PINN training.

3.4. Reservoir connectivity identification

In this section, we discuss the model interpretation of the proposed PINN method. As explained in the previous section, our PINN framework can provide dynamic allocation factors. When we calculate the physics-based residual terms, which are based on material balance equations, the flux terms are computed for every timestep for all connections (Eqs (9)–(11)). We calculate the averaged flux allocation for all timesteps, and it can be visualized as a reservoir connectivity map as shown in Fig. 14. The thickness of the line represents the flux allocation, meaning that thicker line has more flux. Based on the estimated flux allocation map, we can quantitatively analyze which well pair has strong or weak communication, and it helps engineers optimize the field operations. For comparison purposes, we use streamline-based flux allocation as reference. For the streamline calculation, we first run the full-physics

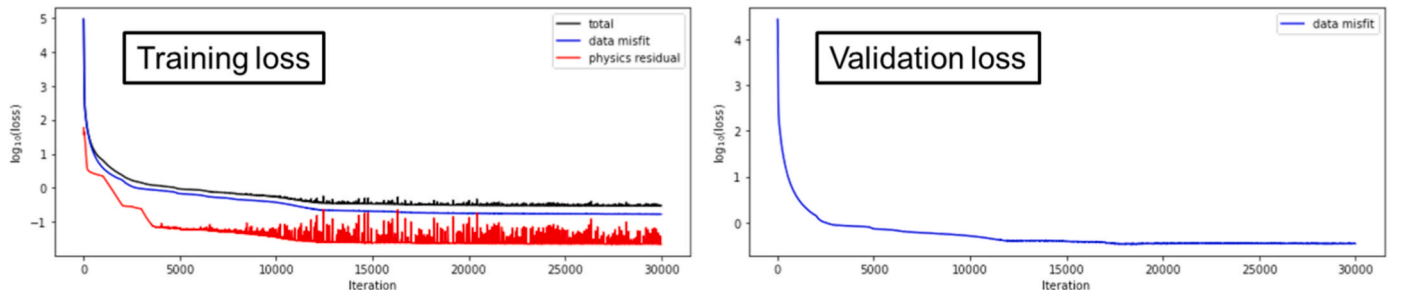


Fig. 22. Training and validation data loss trend during the PINN training for the field case.

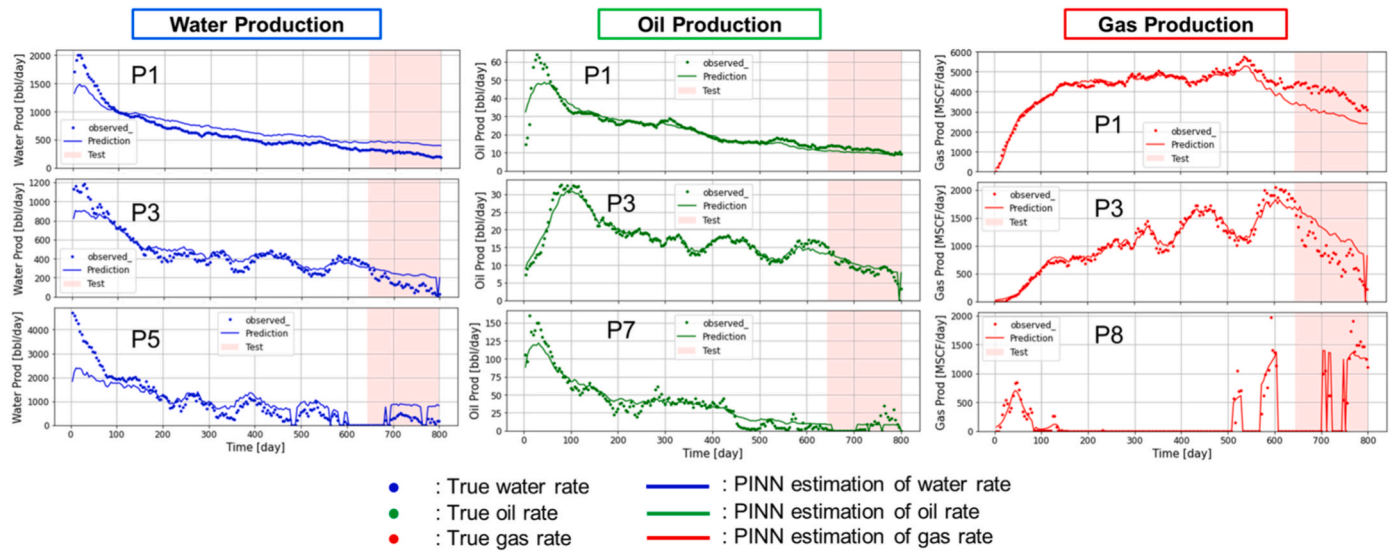


Fig. 23. Training and prediction results of multiphase production rates for selected wells for the field case.

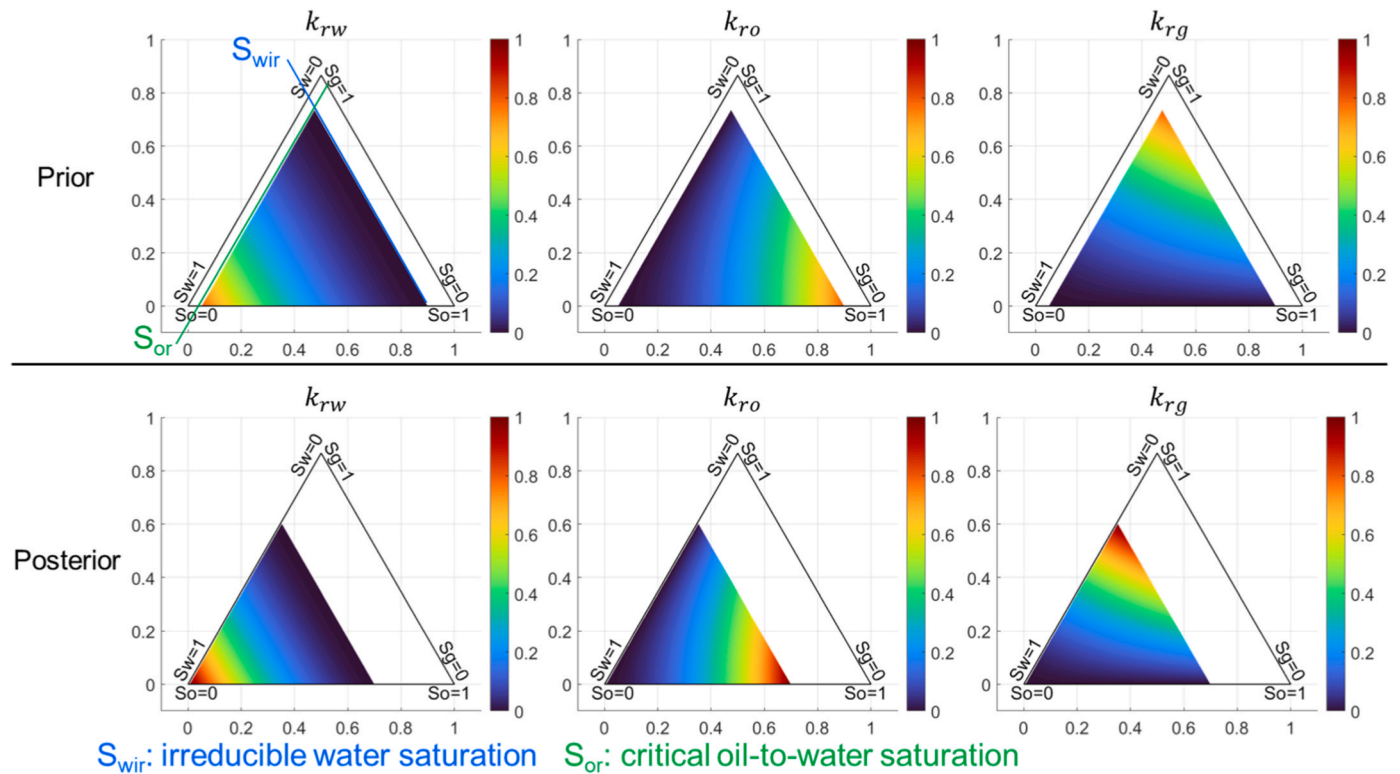


Fig. 24. Three-phase relative permeability models of prior and posterior models for field application.

numerical simulation and obtain the velocity field for all grid cells for all timesteps. Based on the simulated velocity field, we can trace streamlines for all timesteps (Datta-Gupta and King, 2007). Since each streamline carries the same amount of fluid by design, we can calculate the flux allocation between producers and injectors based on the number of streamlines between those two points. The streamlines and the reservoir connectivity map given from streamlines are shown in Fig. 15. Each color of the streamlines represents the streamlines starting from different injectors. Fig. 16 shows the comparison of the strong reservoir

connectivities, larger than flux of 500 [bbl/day], identified from streamline-based flux allocation and PINN framework. Good agreement is confirmed between these two connectivity maps, indicating that PINN can identify reservoir connectivity accurately and solely based on the historical well measurements with orders of magnitude computational speed up. The mean absolute percent error (MAPE) of averaged PINN-based flux allocation over time is calculated against the streamline-based flux allocation, and it is 3.7 % for overall connections.

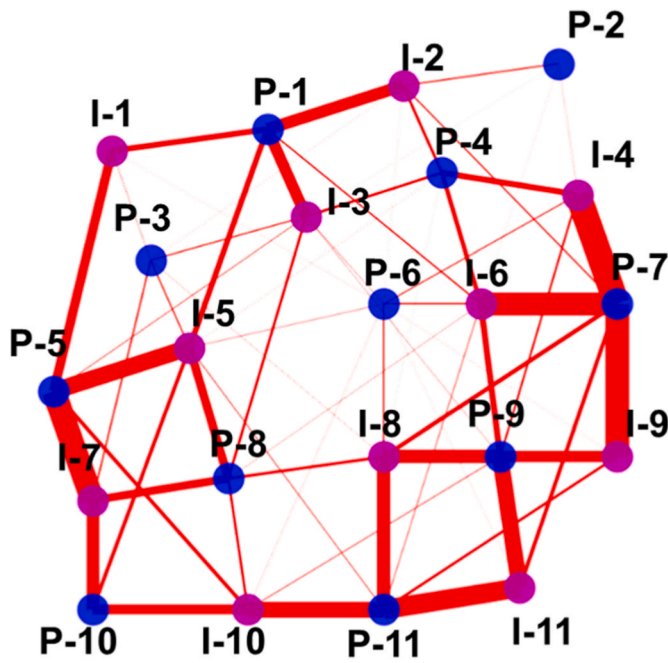


Fig. 25. Reservoir connectivity map from PINN with reduced physics model for the field case.

4. Field application

In this section, the proposed PINN framework is applied to a real field case introduced by Olalotiti-Lawal et al. (2019) and Olalotiti-Lawal et al. (2019), which is a large-scale post-combustion CO₂ injection project for the purpose of EOR and CO₂ storage in the reservoir. We extract a portion of the reservoir model to test the proposed PINN framework.

4.1. Model description

Figs. 17 and 18 shows the reservoir model for the field application. The model has 37x35x101 grids and the domain size is approximately 3680x3272x151 [ft]. Unlike the synthetic reservoir case, the model exhibits a high degree of areal and vertical heterogeneity including impermeable baffles. The depth of the top layer is about 6000 [ft]. The initial saturation and reservoir pressure condition is shown in Figs. 19 and 20. The model has 11 producers and 11 injectors that are completed for the entire vertical zone of the reservoir. The well response data is created by commercial simulation (Schlumberger, 2022), where the injectors are operated under rate constraint and the producers are operated with BHP constraint. As opposed to the synthetic case, some of the producers have well shut-ins. Total 640 days of historical well measurements are available from all the wells, and we conduct future prediction of multiphase production rate for following 160 days. Some sample well measurements are shown in Fig. 21.

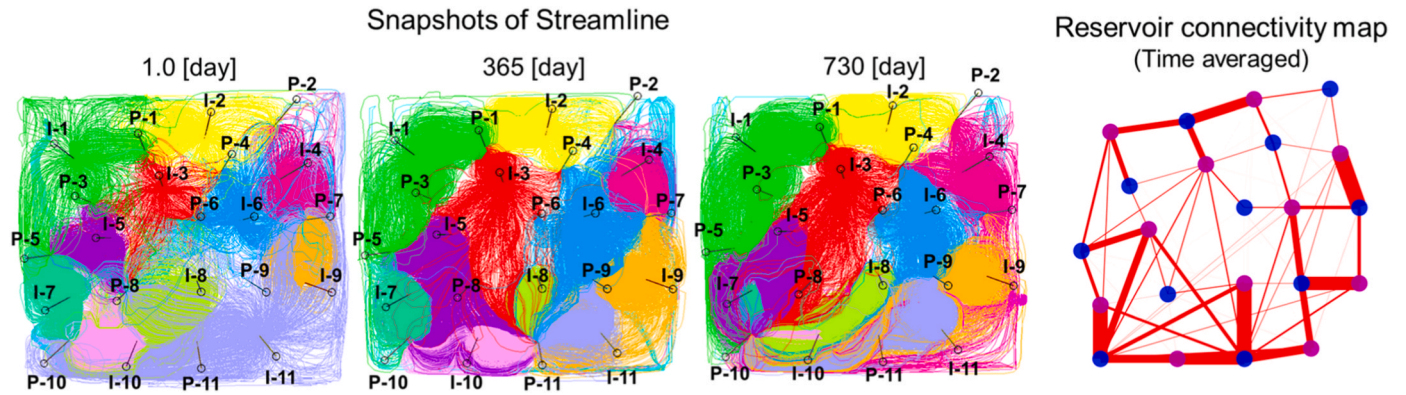


Fig. 26. Streamline tracing map (at selected timesteps) and reservoir connectivity map based on the time averaged flux allocation by streamline for the field case.

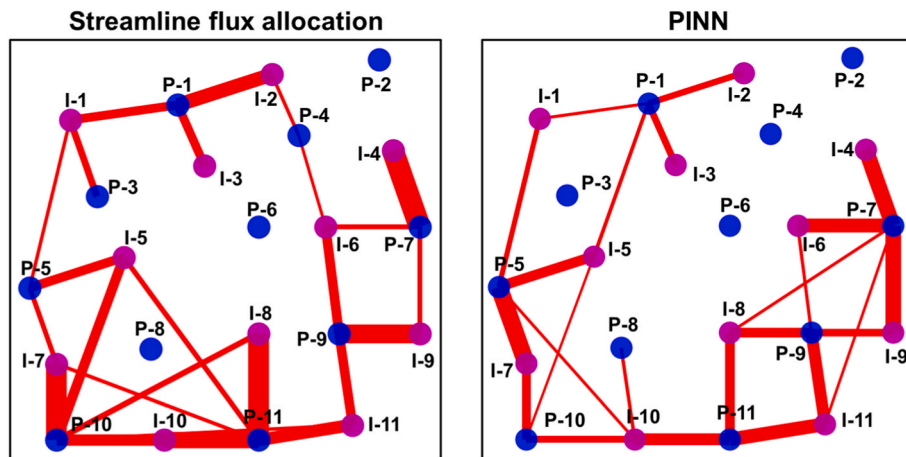


Fig. 27. Comparison of reservoir connectivity maps between streamline-based method and PINN for the field case. Only strong connectivity lines (over 300 [bbl/day]) are shown.

4.2. Model training

The hyperparameters and neural network architecture are described in Tables 13 and 14. Fig. 22 shows that the trend of the training and validation data loss decreases during the PINN training. Both training loss and validation loss decrease with training iterations and stay flat at the end, which indicates that the PINN training is properly conducted without overfitting. The model calibration took about 45 min using the proposed PINN framework. If we use the traditional history matching algorithm such as genetic algorithm with serial runs of full-physics numerical simulation, it will take orders of magnitude more CPU time for this application considering that a single full-physics simulation takes around 100 min.

4.3. Prediction result

Fig. 23 shows the training and prediction performance of the proposed PINN models for three phase production rates for selected three wells. The pink window represents the prediction part. The PINN framework gives reasonable prediction performance for this application. The proposed PINN framework utilizes customized reduced physics model for residual calculation, in which the transient productivity and dynamic allocation factors caused by well shut-ins or workover can be considered properly. The minimization of the physics-based residual helps the neural network avoid overfitting the training production data, enabling stable and physically realistic forecasting of the multiphase production data.

The relative permeability model is calibrated through PINN training, and the prior and posterior models are shown in Fig. 24. Similar to the previous synthetic case, the end point relative permeability of both water phase and hydrocarbon phase increased, and critical oil-to-water saturation is increased. As a result, mobility of hydrocarbon phase increased after model calibration by PINN training.

4.4. Reservoir connectivity identification

The reservoir connectivity is estimated using the proposed PINN framework and compared to the reference reservoir connectivity map created by streamline-based flux allocation that relies on full-physics numerical simulation. Fig. 25 shows the reservoir connectivity map created by the proposed PINN algorithm. The thickness of the line represents the magnitude of the flux allocation. Snapshots of the traced streamlines and the reservoir connectivity map given by streamlines are shown in Fig. 26. Each color of the streamlines represents the streamlines starting from different injectors. Fig. 27 shows a comparison of the strong reservoir connectivities, larger than flux of 300 [bbl/day], identified from streamline-based flux allocation and PINN framework. Reasonable agreement is confirmed between these two connectivity maps, indicating that PINN successfully identified the reservoir connectivity solely based on historical well measurements with reasonable accuracy, although the heterogeneity of the reservoir model is significantly higher than the previous synthetic example. The mean absolute percent error (MAPE) of averaged PINN-based flux allocation over time is calculated against the streamline-based flux allocation, and it is 9.4 % for all connections.

5. Conclusion

- Two novel hybrid models have been developed in this study, which infuse physical laws into neural network architecture for reservoir

connectivity identification and robust multiphase production forecasting in CO₂-EOR problems. The first method is physics-based data preprocessing, and the second method is the physics-informed neural network (PINN) with a reduced physics model.

- The proposed hybrid models are shown to outperform existing pure machine learning model (LSTM) in terms of long-term three-phase production forecasting for the synthetic case application. The physics-based data preprocessing method utilizes CRM solutions as improved input feature of the neural network model, which substantially enhanced the prediction accuracy. The PINN framework exploits customized reduced physics model to efficiently calculate the physics-based regularization terms. Furthermore, the reduced physics model in PINN can account for transient productivity index and dynamic allocation factor, which enhances the prediction capability.
- The proposed PINN framework provides reservoir connectivity in terms of dynamic flux allocation, and this capability is demonstrated for both synthetic case and field application. We obtained good agreement in terms of reservoir connectivity characterization between PINN and full-physics streamline-based flux allocation for both synthetic case and field case applications. The reservoir connectivity directly informs production operation management actions such as optimizing injection patterns and injection allocation optimization for mitigating early breakthrough of injected fluid. Although the field case reservoir has highly heterogeneous geologic property distribution, the proposed PINN framework showed robust estimation performance in terms of reservoir connectivity without relying on geological knowledge.
- The proposed hybrid models are calibrated solely based on the historical well measurements without relying on the prior geological information. Very often it is challenging to obtain reliable geological models and history matching using full physics numerical simulations can be computationally prohibitive especially for CO₂-EOR reservoirs because the full-physics simulation involves three phase and multi-component fluid behavior with miscible displacement. The proposed hybrid models can overcome these challenges.
- The proposed hybrid models provide more stable training and robust prediction performance compared to pure machine learning models because they incorporate physics into the architecture, preventing the model from overfitting to noisy data.
- Although the proposed hybrid models showed superior performance prediction, there is still room for the improvement of the production forecasting. There are several potential ideas that can be pursued for this purpose. First, we can use recurrent neural networks such as LSTM and GRU for the hybrid models instead of feed-forward neural networks. Second, more complex physics can potentially be considered in the customized reduced physics model. In the proposed PINN framework in this study, we have a single pore volume or tank for each well, which could be too simplistic to properly capture the complex physics involving three phase miscible displacement. Instead, we can create multiple tanks for each well or for each connection in the system similar to the RGNet (Sankaran and Sun, 2020) or StellNet (Lutidze, 2018).

CRedit authorship contribution statement

Masahiro Nagao: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Akhil Datta-Gupta:** Writing – review & editing,

Supervision, Project administration, Funding acquisition.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. — Miscible viscosity, density, and three-phase relative permeability calculation

This section describes the viscosity, density, and relative permeability calculation using in the proposed PINN framework. The described models in this section are all based on the technical description by [Schlumberger \(2022\)](#).

Viscosity model

The effective oil and gas viscosities are given by:

$$\mu_{o\text{ eff}} = \mu_o^{1-\omega} \mu_m^\omega \quad (\text{A.1})$$

$$\mu_{g\text{ eff}} = \mu_g^{1-\omega} \mu_m^\omega \quad (\text{A.2})$$

where ω is the Todd-Longstaff mixing parameter lies between 0 and 1. The mixture viscosity μ_m is given by:

$$\mu_m = \frac{\mu_o \cdot \mu_g}{\left(\frac{S'_g}{S'_n} \mu_o^{0.25} + \frac{S'_o}{S'_n} \mu_g^{0.25} \right)^4} \quad (\text{A.3})$$

$$S'_o = S_o - S_{or}, \quad S'_g = S_g - S_{gc}, \quad S'_n = S'_o + S'_g \quad (\text{A.4})$$

where S_{or} is the critical oil saturation, which is the highest oil saturation where oil phase be immobile, and S_{gc} is critical gas saturation, which is the highest gas saturation where gas phase be immobile. In the proposed PINN framework, ω can be one of the calibration parameters.

Density model

The miscible density is calculated as follows. First the partially mixed or effective viscosities are calculated using above formula. The effective viscosity is used to calculate an effective saturation fraction to be used in oil and gas density calculation.

$$\left(\frac{S_o}{S_n} \right)_{oe} = \frac{\mu_o^{0.25} (\mu_{o\text{ eff}}^{0.25} - \mu_g^{0.25})}{\mu_{o\text{ eff}}^{0.25} (\mu_o^{0.25} - \mu_g^{0.25})} \quad (\text{A.5})$$

$$\left(\frac{S_o}{S_n} \right)_{ge} = \frac{\mu_o^{0.25} (\mu_{g\text{ eff}}^{0.25} - \mu_g^{0.25})}{\mu_{g\text{ eff}}^{0.25} (\mu_o^{0.25} - \mu_g^{0.25})} \quad (\text{A.6})$$

The effective oil and gas densities are calculate using pure component densities as follows:

$$\rho_{o\text{ eff}} = \rho_o \left(\frac{S_o}{S_n} \right)_{oe} + \rho_g \left[1 - \left(\frac{S_o}{S_n} \right)_{oe} \right] \quad (\text{A.7})$$

$$\rho_{g\text{ eff}} = \rho_o \left(\frac{S_o}{S_n} \right)_{ge} + \rho_g \left[1 - \left(\frac{S_o}{S_n} \right)_{ge} \right] \quad (\text{A.8})$$

Miscible three-phase relative permeability model

The injected gas and reservoir oil are considered to be miscible components of the hydrocarbon phase. The flow is two-phase in character and two-phase relative permeability curves are given as follows:

$$k_{rw} = k_{rw} E \cdot S_w^{nw} \quad (\text{A.9})$$

$$k_m = k_{rn} E \cdot S_n^{nn} \quad (\text{A.10})$$

$$S'_w = \frac{S_w - S_{wc}}{1 - S_{gc} - S_{or} - S_{wc}} \quad (\text{A.11})$$

$$S'_n = \frac{S_n - S_{gc} - S_{or}}{1 - S_{gc} - S_{or} - S_{wc}} \quad (\text{A.12})$$

where $k_{rw}E$ is the end point water phase relative permeability, $k_{rn}E$ is the end point hydrocarbon phase relative permeability, S_{wc} is connate water saturation, S_{or} is critical oil-to-water saturation, S_{gc} is critical oil-to-gas saturation, and $S_n = S_o + S_g$. Then, the oil and gas phase relative permeabilities are computed as saturation weighted fraction as follows:

$$k_{rg} = \left(\frac{S_g - S_{gc}}{S_n - S_{gc} - S_{or}} \right) k_{rn}(S_n) \quad (\text{A.13})$$

$$k_{ro} = \left(\frac{S_o - S_{or}}{S_n - S_{gc} - S_{or}} \right) k_{rn}(S_n) \quad (\text{A.14})$$

In the proposed PINN framework, the relative permeability model can be calibrated by changing $k_{rw}E$, nw , $k_{rn}E$, nn , S_{wc} , S_{or} , and S_{gc} .

Appendix B. — Penalty coefficient tuning for reservoir connectivity identification

In this section, we describe the strategy to optimize the penalty coefficient w_{dist} in Eq. (25) for reservoir connectivity identification. In the synthetic case application, we trained the PINN model using different w_{dist} values and recorded the training MSE based on Eq. (25). The reservoir connectivity map and training MSE obtained are provided in Fig. A.1. As you increase the penalty coefficient, the training MSE stays flat first ($w_{dist} = 0.001$ to $w_{dist} = 0.1$), and then it starts to increase. Compared to the connectivity map given by streamline flux allocation, the PINN connectivity maps of $w_{dist} = 0.01$ and $w_{dist} = 0.1$ shows many spurious reservoir connections. As you increase the penalty coefficient, the spurious connections disappear without significant increase of train MSE. Therefore, the optimal penalty coefficient can be determined as point where the training MSE starts to increase. Based on this analysis, we can find the optimal penalty coefficient, and it was chosen to be 0.8 for the synthetic case. The same process is conducted for the field application, and the results are shown in Fig. A.2.

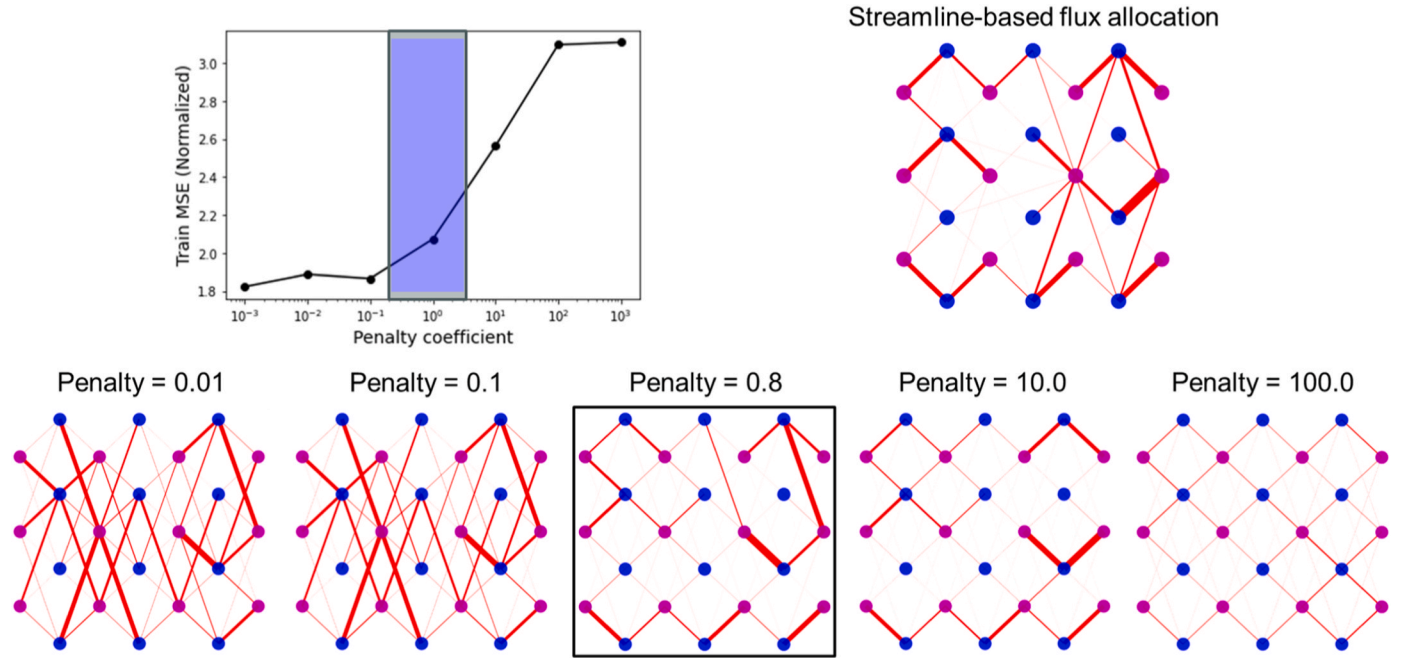


Fig. A.1. Result of the penalty coefficient tuning for the synthetic case application.

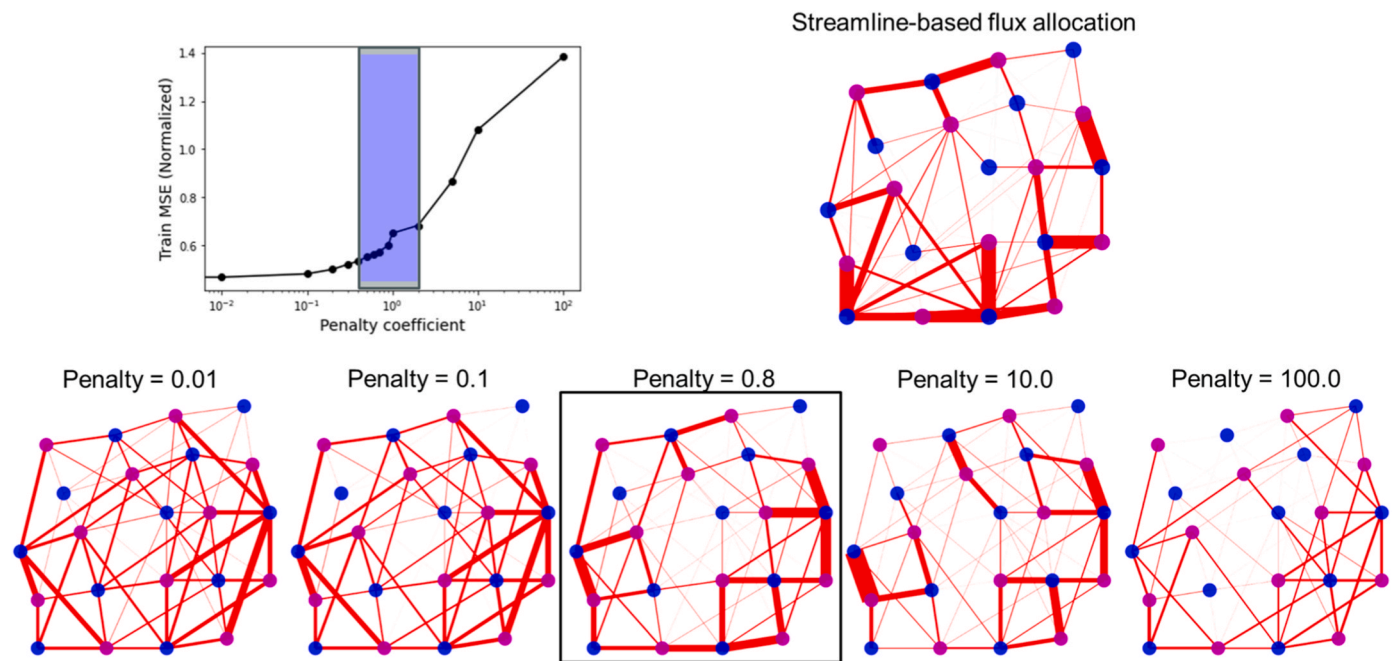


Fig. A.2. Result of the penalty coefficient tuning for the field case application.

Data availability

The authors do not have permission to share data.

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